TOPIC 01

Foundations

Population, sample, parameter, statistic and the logic of inference

WHAT YOU WILL LEARN

- Tell descriptive and inferential statistics apart
- Distinguish a population from a sample, and a parameter from a statistic
- Recognise the main types of data and the main sampling methods
- Understand why samples differ from each other, and why biased samples cannot be rescued
- See the whole inference workflow that the rest of this handbook builds

1. Why statistics?

Data is rarely complete. A company cannot ask every customer what they think, a doctor cannot test a drug on every patient in the world, and a factory cannot destroy every bulb to learn how long bulbs last. We almost always see **part** of the data and need to say something about the **whole**. Statistics gives us the rules for doing that honestly, including how unsure we should be.

The two branches

	Descriptive statistics	Inferential statistics
Question answered	What does *this* data look like?	What can we say about the larger group the data came from?
Tools	Mean, median, mode, range, variance, standard deviation, charts	Probability, distributions, estimation, confidence intervals, hypothesis tests
Output	Summaries that are exact for the data in hand	Statements with uncertainty attached (a range, a probability)
Example	The average expense of the 1,000 households in our file is 18,000	The average expense of *all* households in the city is probably between 17,200 and 18,800

You have already met descriptive statistics. Everything in this handbook is about the second column: using probability to move from a **sample** to a **population**.

2. Population and sample

DEFINITION **Population**

The complete set of all individuals, items or measurements we want to learn about. It is defined by the question, not by convenience: 'all adults in India', 'all bulbs this factory will ever produce', 'all visits to our website'.

DEFINITION **Sample**

A subset of the population that we actually observe. The number of items in it is the sample size, written n. The population size is written N.

A number that describes a population is a **parameter**. A number computed from a sample is a **statistic**. Parameters are fixed but usually unknown. Statistics are known but change from sample to sample. We use the statistic to **estimate** the parameter.

Quantity	Population (parameter)	Sample (statistic)
Size	N	n
Mean	μ (mu)	$\bar{x}$ (x-bar)
Standard deviation	σ (sigma)	s
Variance	σ^2	s^2
Proportion	p (or π)	$\hat{p}$ (p-hat)
Correlation	ρ (rho)	r

Note that the sample standard deviation s divides by **n − 1**, not n; the reason is explained in Topic 08.

> **KEY IDEA** **Memory trick**
> Greek letters mostly describe the **population** (the truth we cannot see) and Roman letters the **sample** (what we can compute). The proportion is the one exception: p is the population value and $\hat{p}$ the sample value.

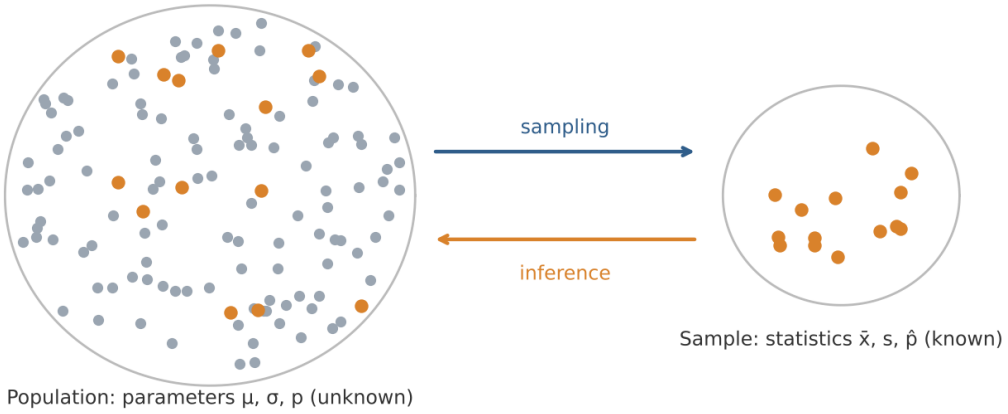


Figure 1. Sampling moves from the population to a sample; inference moves back from the sample to the population.

> **NOTE** **See also**
> The descriptive-statistics notes already cover population versus sample in detail (folder `02_Population_vs_Sample`). This section is a short inference-focused recap.

3. Types of data

The kind of data decides which summaries and tests make sense, so it is worth naming the types.

Type	Sub-type	What it means	Examples
Categorical (qualitative)	Nominal	Labels with no order	Blood group, city, gender
	Ordinal	Labels with a natural order	Education level, satisfaction (low/medium/high)
Numerical (quantitative)	Discrete	Counted, whole numbers	Number of children, defects per batch
	Continuous	Measured, any value in a range	Height, time, income, temperature

> **NOTE** **Recap**
> This is a short recap of the descriptive-statistics folder `03_Types_of_Variables`, which covers categorical and numerical variables in full.

> **NOTE**
> Averages are meaningful for numerical data. For categorical data we use counts and **proportions** instead, and that is why proportions get their own formulas later in this handbook.

4. How samples are chosen

Inference only works if the sample is **representative**. The best protection is to let chance, not people, decide who is selected.

Method	How it works	Strength	Weakness
Simple random	Every member has an equal chance; draw like names from a hat	Unbiased, the basis of most formulas	Needs a full list of the population
Systematic	Every k-th member after a random start	Easy to carry out	Bad if the list has a hidden pattern
Stratified	Split into groups (strata), sample randomly inside each	Guarantees every group is represented; lower variance	Needs group information beforehand
Cluster	Randomly choose whole groups (schools, villages) and survey everyone in them	Cheap when the population is spread out	Less precise than simple random
Convenience	Whoever is easy to reach	Fast and cheap	Almost always biased
Voluntary response	People choose to respond (online polls)	Easy	Strongly biased towards people with strong opinions

Only the first four are **probability samples**. The formulas in this handbook assume simple random sampling unless stated.

5. Variation and bias: two different problems

Two samples from the same population will give different statistics. That natural difference is **sampling variability** (or sampling error). It is not a mistake, it is chance, and statistics can measure it and shrink it by taking larger samples.

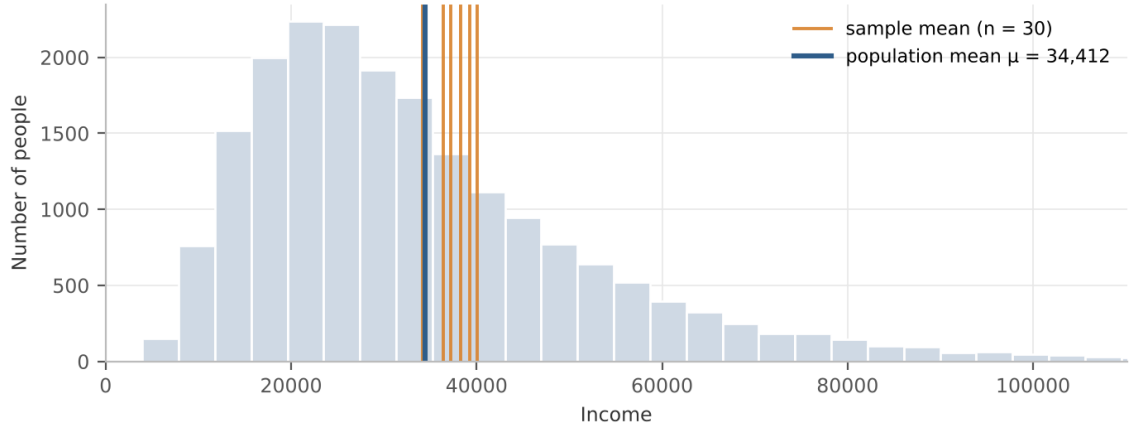


Figure 2. A skewed population of 20,000 incomes. Six random samples of 30 each give a different mean, but they scatter around the true mean μ.

In Figure 2 the six sample means are 36,433, 37,167, 34,169, 40,069, 39,268, 38,312, against a true mean of 34,412. None is exactly right, yet none is wildly off, and the spread between them is something we can quantify. That is the subject of Topics 05 to 07.

COMMON MISTAKE **Bias is different and much worse**

Bias is a *systematic* error that pushes every sample the same way. A bigger sample does not cure it; it just gives you a more precise wrong answer.

- **Selection bias**: the way you choose people favours some types (surveying only people at a gym about exercise habits).
- **Non-response bias**: the people who do not answer differ from those who do.
- **Survivorship bias**: only the 'survivors' are in the data (studying only successful start-ups).
- **Measurement bias**: the question or instrument pushes answers in one direction.

WORKED EXAMPLE **The 1936 Literary Digest poll**

The magazine mailed ballots to millions of people chosen from telephone and car-owner lists and got over two million replies. It predicted that Landon would beat Roosevelt in a landslide. Roosevelt won easily. The huge sample was drawn from wealthier households (in the Depression, not everyone had a phone or car) and responses were voluntary, so it was heavily biased. A far smaller *random* sample would have done better.

6. The inference workflow

Every inferential analysis follows the same chain. The topics in this handbook are the links.

Step	What happens	Where you learn it
1	Define the question and the population	Topic 01 (this one)
2	Draw a random sample and compute a statistic ($\bar{x}$, $\hat{p}$, ...)	Topics 01, 05
3	Use probability to model chance outcomes	Topics 02, 03
4	Know how the statistic behaves from sample to sample (its *sampling distribution*: the pattern of values the statistic takes across many possible samples; uses the normal curve)	Topics 04, 05, 06
5	Measure how precise the statistic is (*standard error*: the typical size of the statistic's sampling error)	Topic 07
6	Turn the statistic into an *estimate* (a single best-guess value) of the parameter	Topic 08
7	Report a range of plausible values (*confidence interval*: a range of values likely to contain the true parameter)	Topic 09
8	Or decide between two claims (*hypothesis test*)	Topics 10, 11

7. A first look in Python

```
import numpy as np
rng = np.random.default_rng(0)

population = rng.lognormal(mean=10.3, sigma=0.55, size=20_000)  # everyone
print("True mean (parameter):", population.mean().round())

sample = rng.choice(population, size=30, replace=False)          # a simple random sample
print("Sample mean (statistic):", sample.mean().round())
print("Sample std s:", sample.std(ddof=1).round())  # ddof=1: sample
```

SUMMARY

- Inferential statistics estimates **population parameters** from **sample statistics**.
- Parameters are fixed and unknown (Greek letters); statistics are known and vary (Roman letters).
- Random sampling controls bias; bigger samples reduce *variability* but not bias.
- Every later topic is a step in the chain: probability, distributions, sampling distribution, standard error, estimate, interval, test.

Practice questions

Try each question before reading the answer underneath it.

Q1. A university surveys 200 of its 12,000 students about library use. Identify the population, the sample, n and N.

Answer. Population: all 12,000 students. Sample: the 200 surveyed. n = 200, N = 12,000.

Q2. The average wait time of all callers last year was 4.2 minutes. A sample of 50 callers averaged 4.6 minutes. Which number is the parameter and which is the statistic?

Answer. 4.2 minutes is the parameter (μ, population). 4.6 minutes is the statistic ($\bar{x}$, sample).

Q3. An online poll asks viewers of a news site to vote on a political question and gets 100,000 votes. Why is it not a reliable estimate for the whole country?

Answer. It is a voluntary-response sample from one site's audience, so it is biased. Its large size does not fix bias.

Q4. Is sampling variability the same as bias?

Answer. No. Variability is random difference between samples and shrinks as n grows. Bias is systematic and does not shrink with n.

Q5. Which sampling method guarantees that each of 4 age groups is represented?

Answer. Stratified sampling, with the age groups as strata.